Linear Algebra (MT-1004)

Homework#18

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Question 1

1. Given the vectors $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, verify that they form an orthogonal basis for $\mathbb{R}^2$.
2. Construct an orthogonal basis for the subspace spanned by $\mathbf{a}_1 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$ and $\mathbf{a}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$.
3. Explain why orthogonal bases are numerically stable in computations.
4. **Python Task:** Write a function that checks if a list of vectors form an orthogonal set.

```
def is_orthogonal_set(vectors):
    # Your code here
```